

ITALIAN FINANCIAL CHALLENGE

AI-Powered Financial Health Classification

LUISS Guido Carli - Grand Challenge

March 23, 2026

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AGENDA

1. Challenge Definition

Objective, business context, success metrics

2. Technical Setup & Infrastructure

Panel structure, temporal constraints, anti-leakage controls

3. Exploratory Data Analysis

Class imbalance, survivorship bias, structural patterns

4. Key Findings & Modeling Strategy

Critical insights, feature engineering, final approach

CHALLENGE OVERVIEW



Objective

Classify Italian Companies into
4 financial health categories (A→D)

A = Excellent

B = Good

C = Moderate Risk

D = High Risk/Distressed



Training Instance Definition

Each row = independent diagnostic
snapshot (company, year) → (X, y)



Business Context

Mirrors real-world credit rating systems
used by banks and investors

- Misclassifying D as A/B → loan defaults, portfolio losses
- Recall on Class D is critical business priority



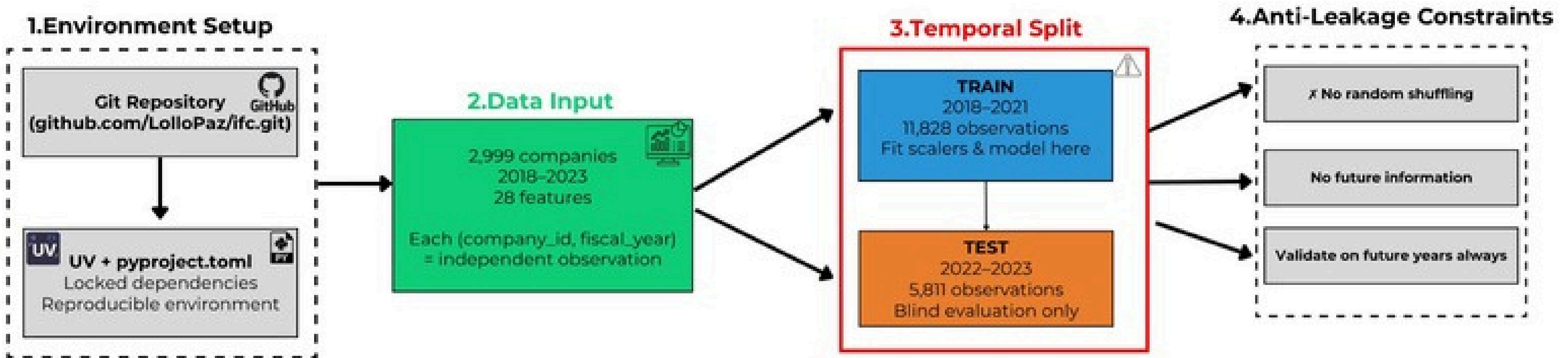
Success Metrics

Primary: Weighted **F1-Score** ≥ **0.65**

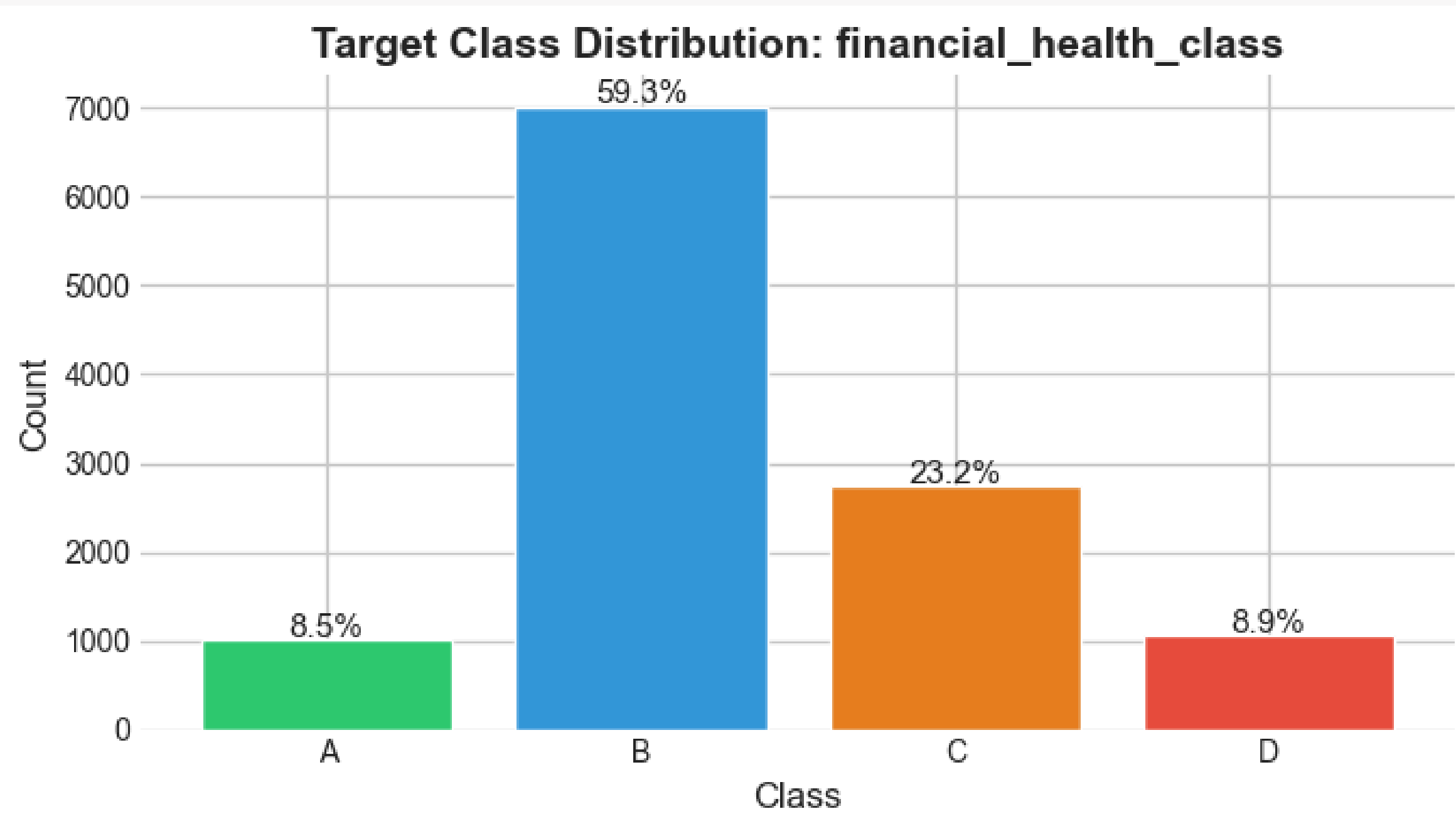
Avoid severe misclassifications (A↔D errors)

Interpretable features for business stakeholders

PROJECT SETUP & TECHNICAL INFRASTRUCTURE



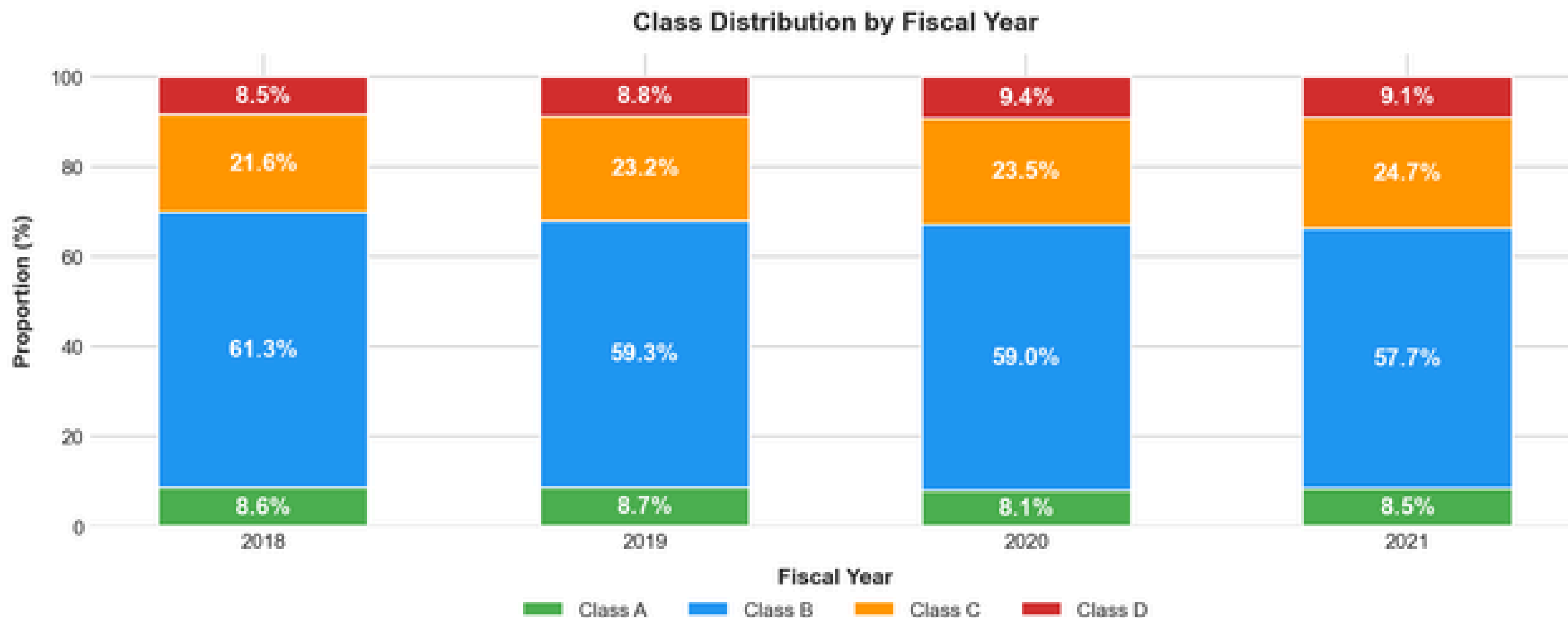
CLASS DISTRIBUTION & IMBALANCE



MODELING IMPLICATION

- Class B dominates at 59.3%
- Weighted F1-Score as primary metric
- `class_weight="Balanced"` strategy

TEMPORAL STABILITY (2018–2021)



Chi² Stability Test

$\chi^2 = 11.64$

$p = 0.234$

dof = 9

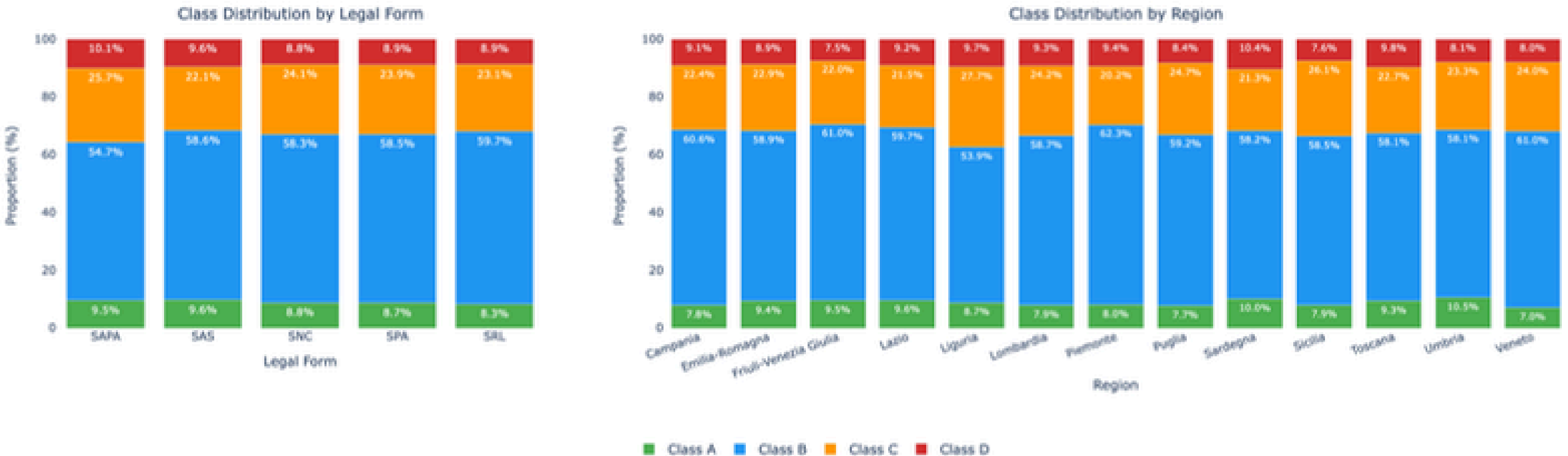
Not Significant

Distribution stable over time

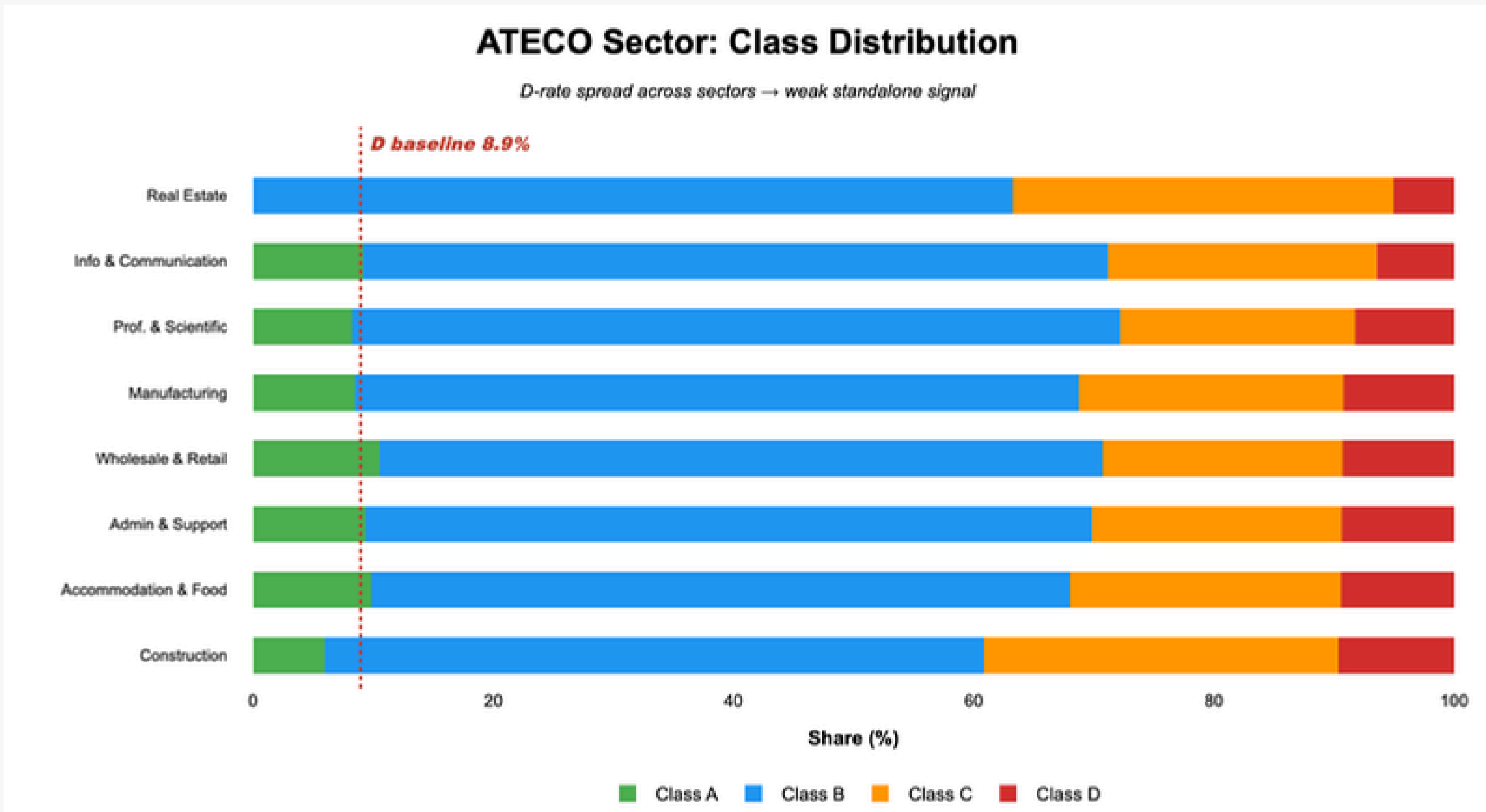
No **COVID-19 spike** detected

CATEGORICAL FEATURES: LOW STANDALONE SIGNAL & STRATEGY

Low-Signal Categoricals: Legal Form & Region



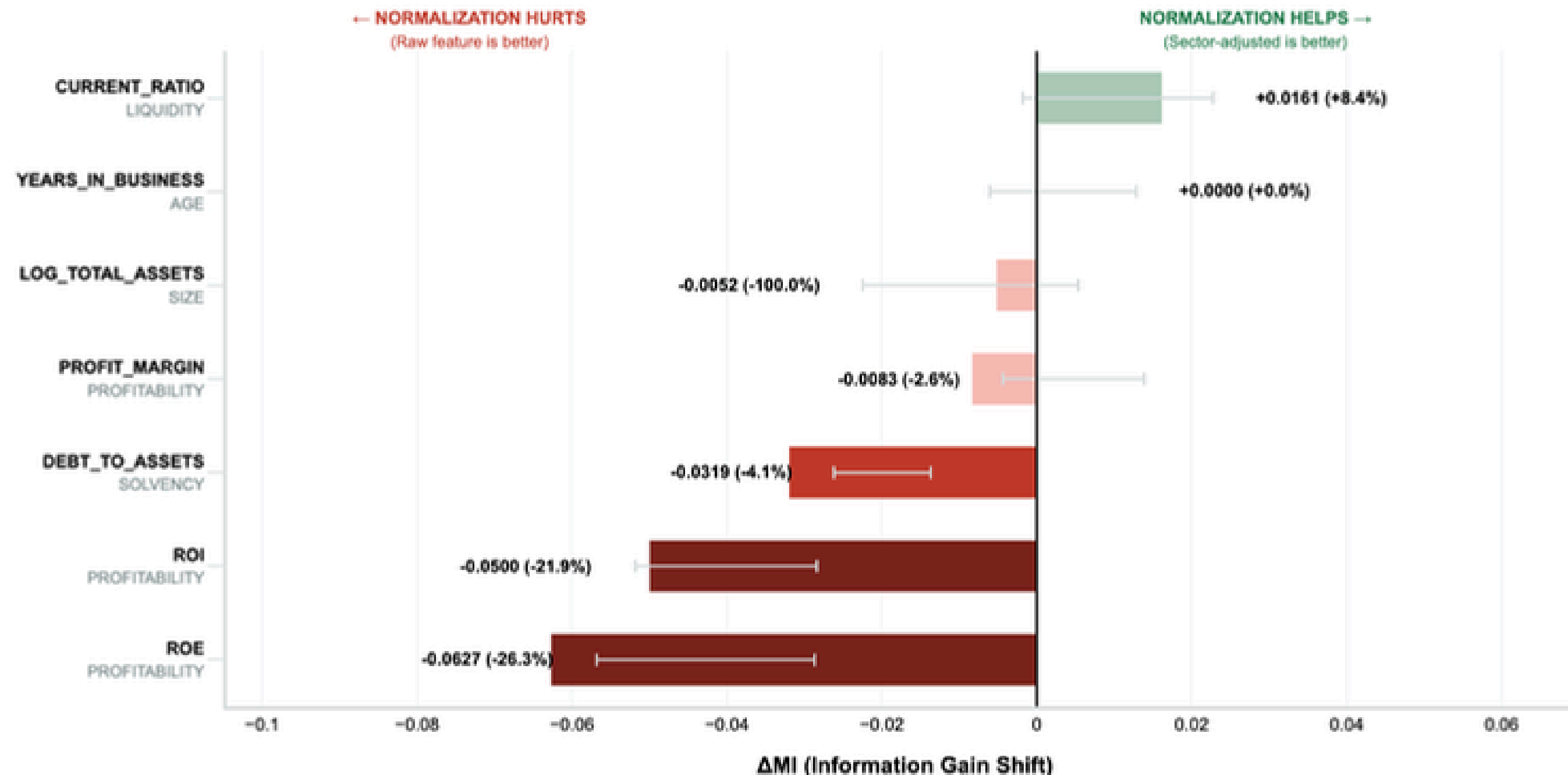
CATEGORICAL FEATURES: LOW STANDALONE SIGNAL & STRATEGY



CATEGORICAL FEATURES: LOW STANDALONE SIGNAL & STRATEGY

Does Sector Normalization Improve Predictive Power?

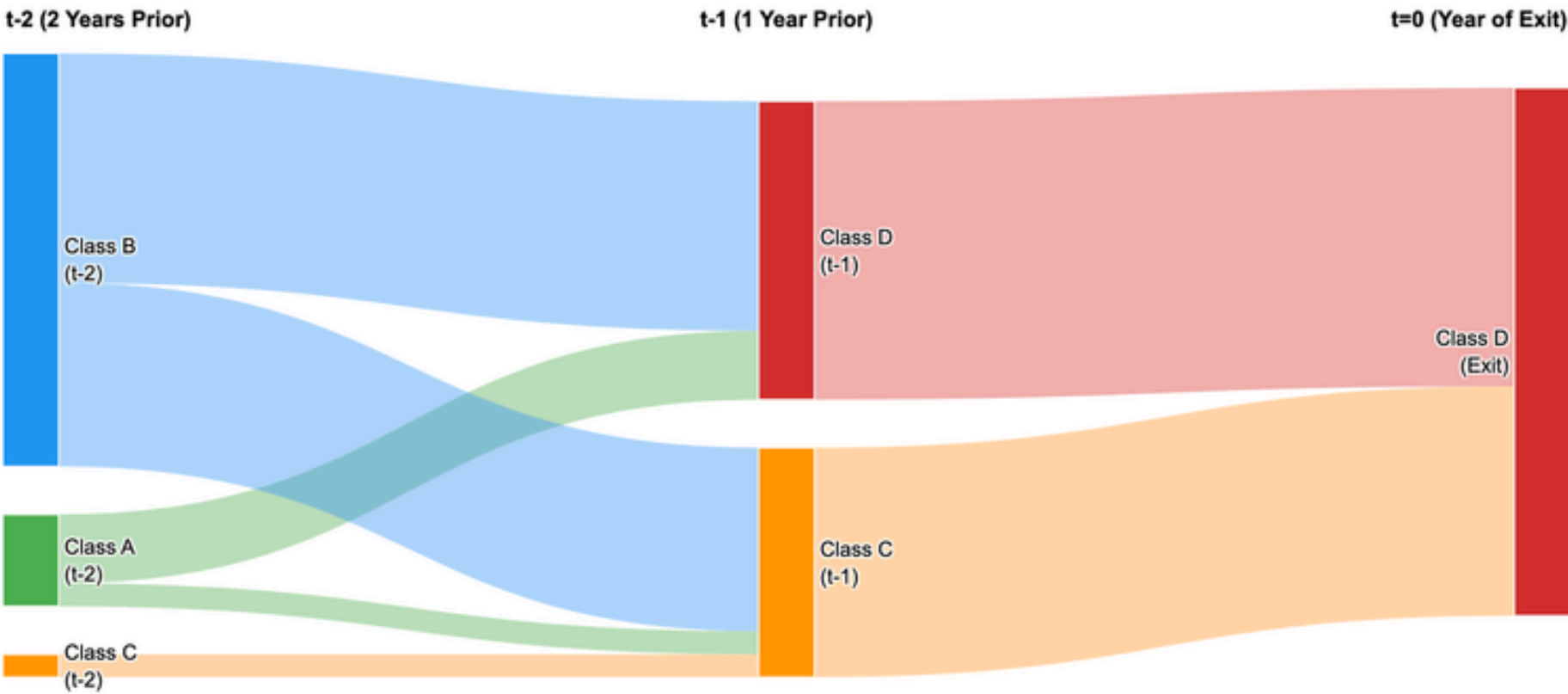
Bars: Δ Information Gain (ΔMI) · Whiskers: 90% Bootstrap Confidence Interval



THE BIAS IS THE FEATURE

Exit Trajectory & Class Transitions

Data-driven flow based on 67 exiting companies. Shows the exact breakdown into Class C before final exit.



- 100% of exits are preceded by a Class D state.
- Distress is a traceable process, not a random event.

Credit Migration Matrix: Year-over-Year Transition Probabilities

n=8,896 transitions · Rows sum to 100% · High-Contrast Intensity Applied

Year t	→ A	→ B	→ C	→ D	→ EXIT
From A	11.0%	60.6%	20.8%	7.6%	—
From B	8.2%	58.4%	24.6%	8.8%	—
From C	8.2%	58.2%	24.2%	9.3%	—
From D	7.6%	54.4%	18.5%	11.1%	8.5%

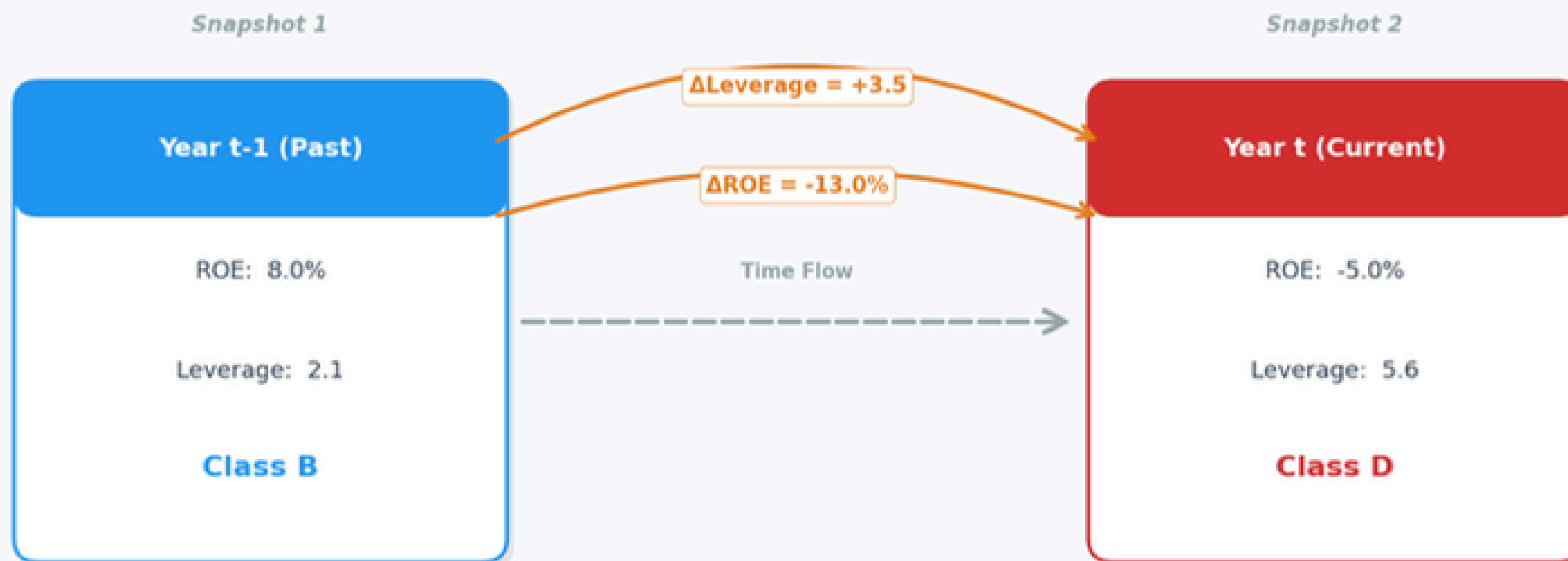
■ Financial Upgrade ■ Credit Stability ■ Financial Downgrade ■ Default / Exit

- Mean Reversion Trap: ~**60% flow to Class B** ~9% drop to Class D
- **8.5% Exit (Row D only)**

THE BIAS IS THE FEATURE

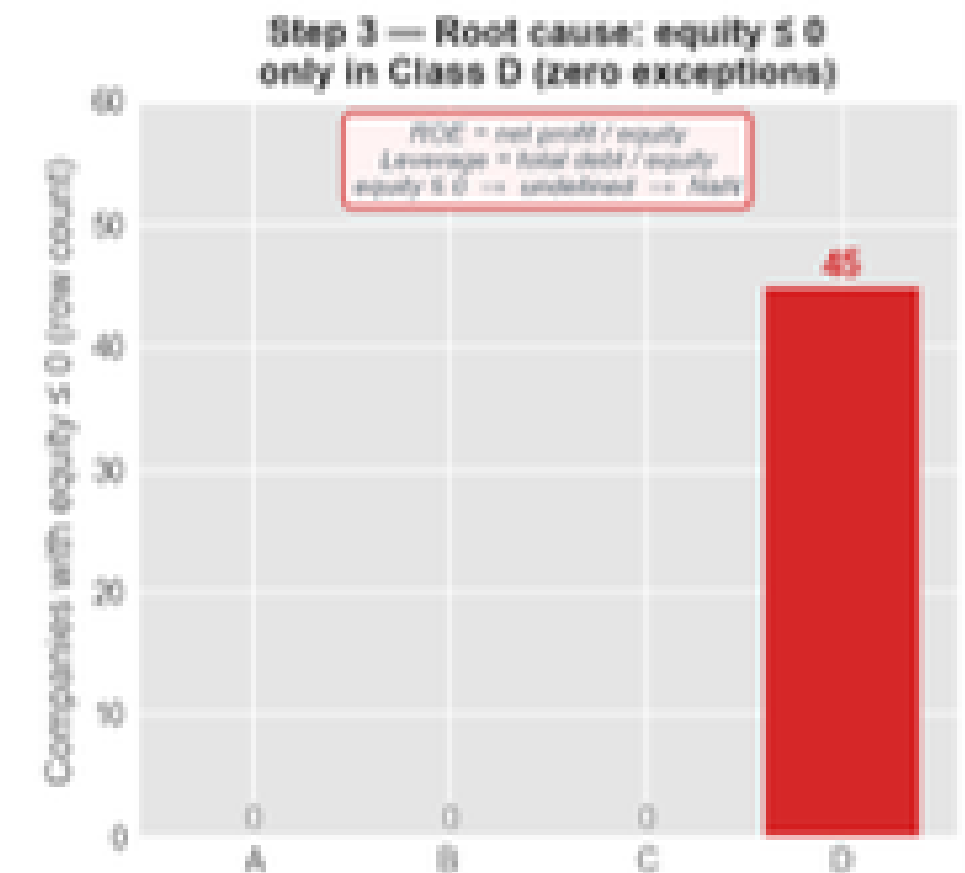
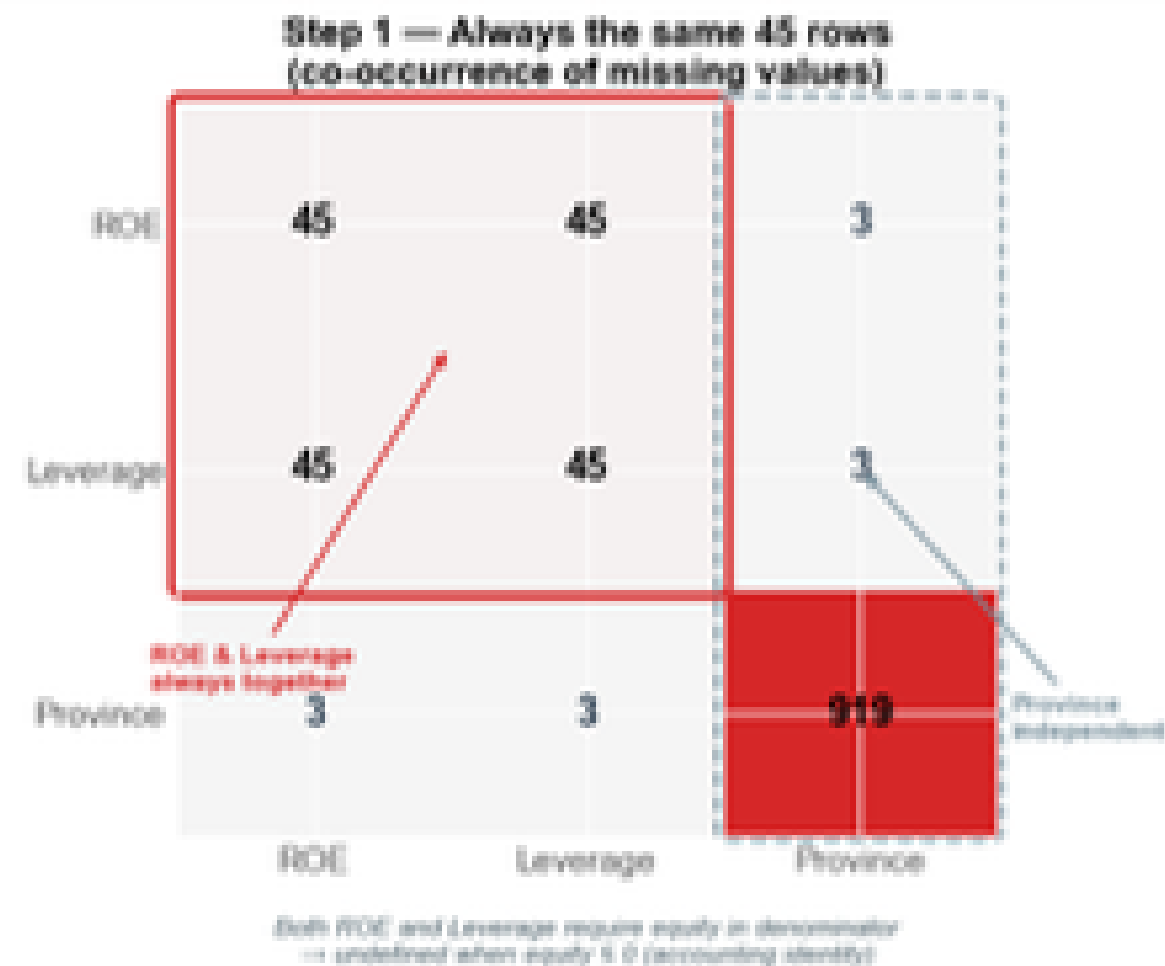
From Static Snapshots to Trajectory Modeling

We engineered YoY changes (Δ) and historical states to capture the momentum of financial deterioration.



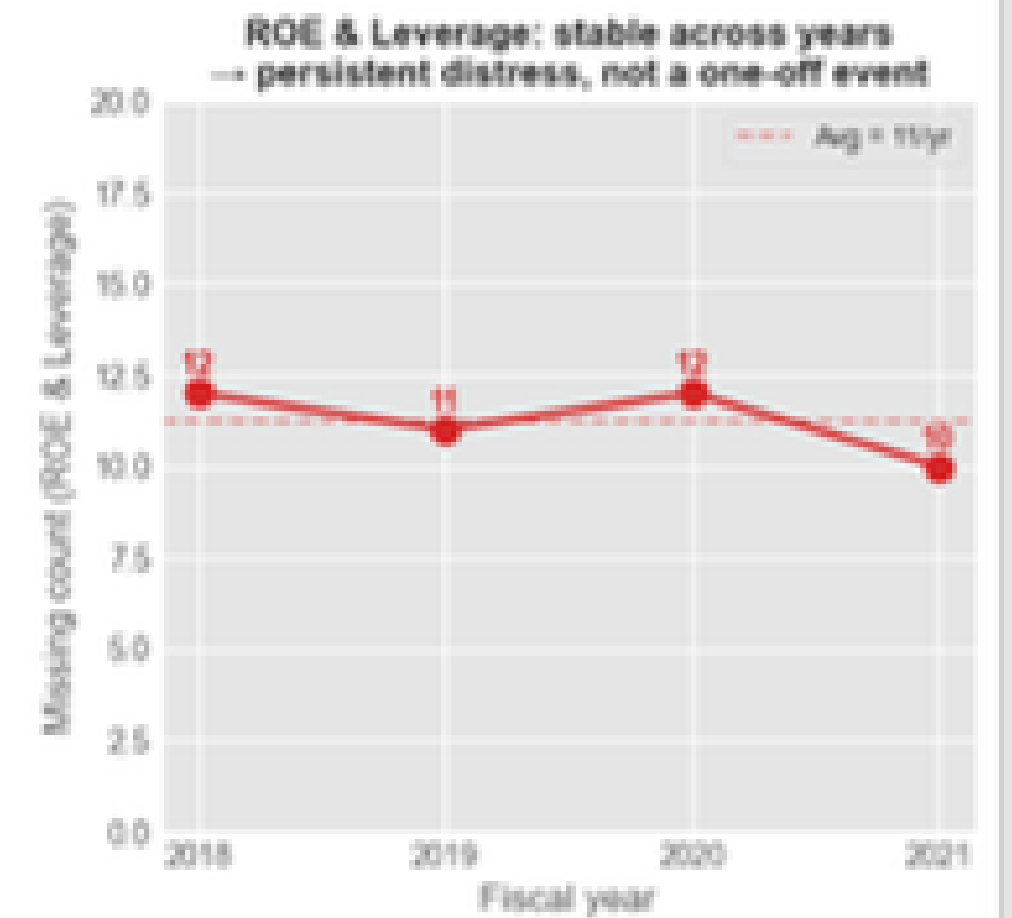
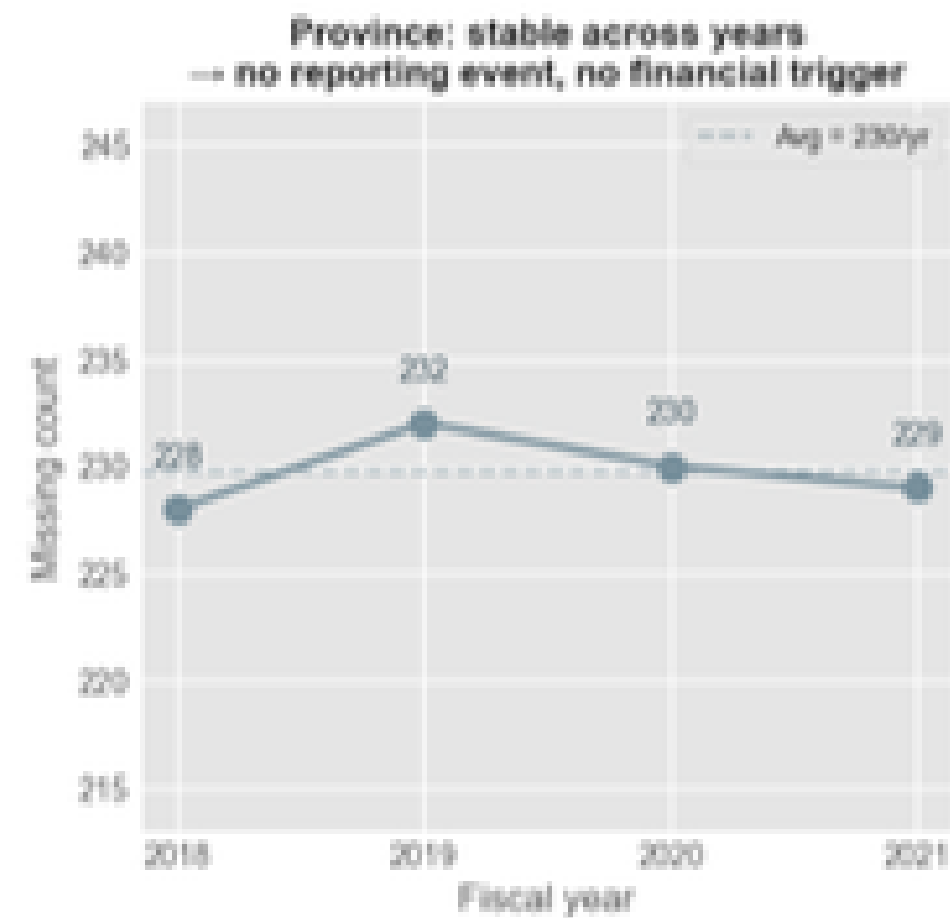
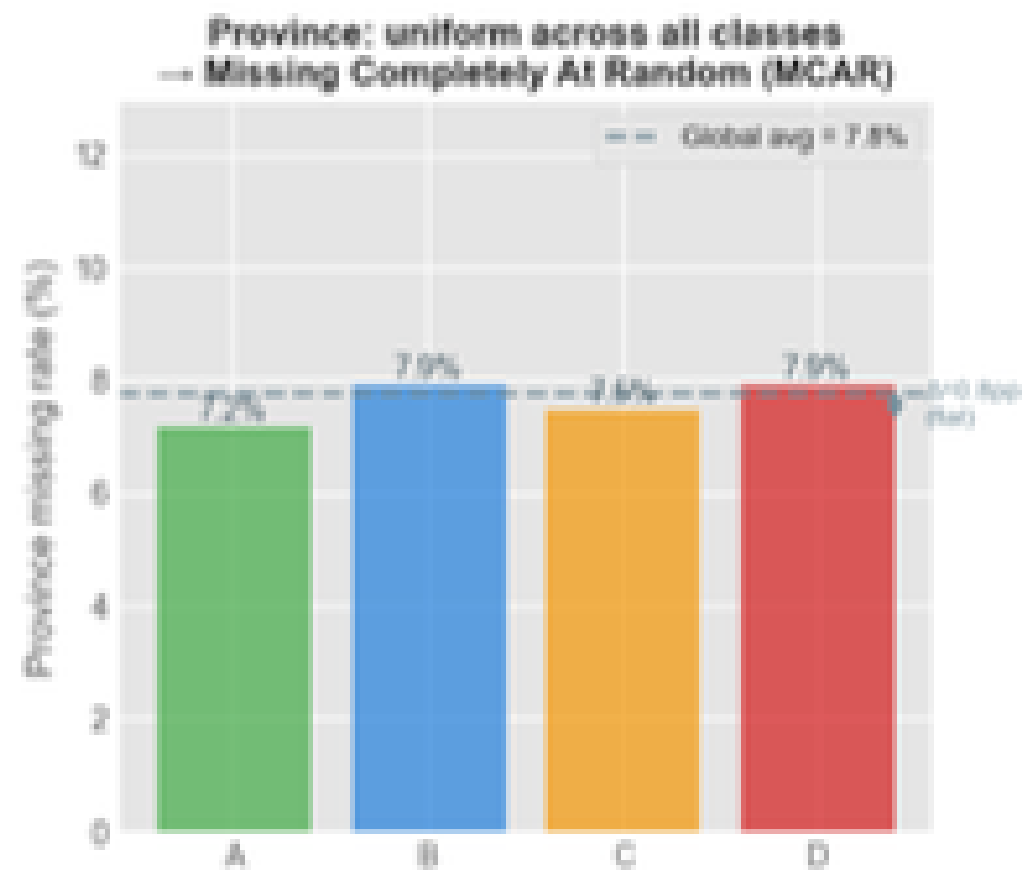
MISSING VALUES: TWO MECHANISMS, TWO RESPONSES

STRUCTURAL MISSINGNESS: equity ≤ 0 $\rightarrow$ ROE & Leverage undefined $\rightarrow$ 100% Class D $\rightarrow$ encode as distress flag

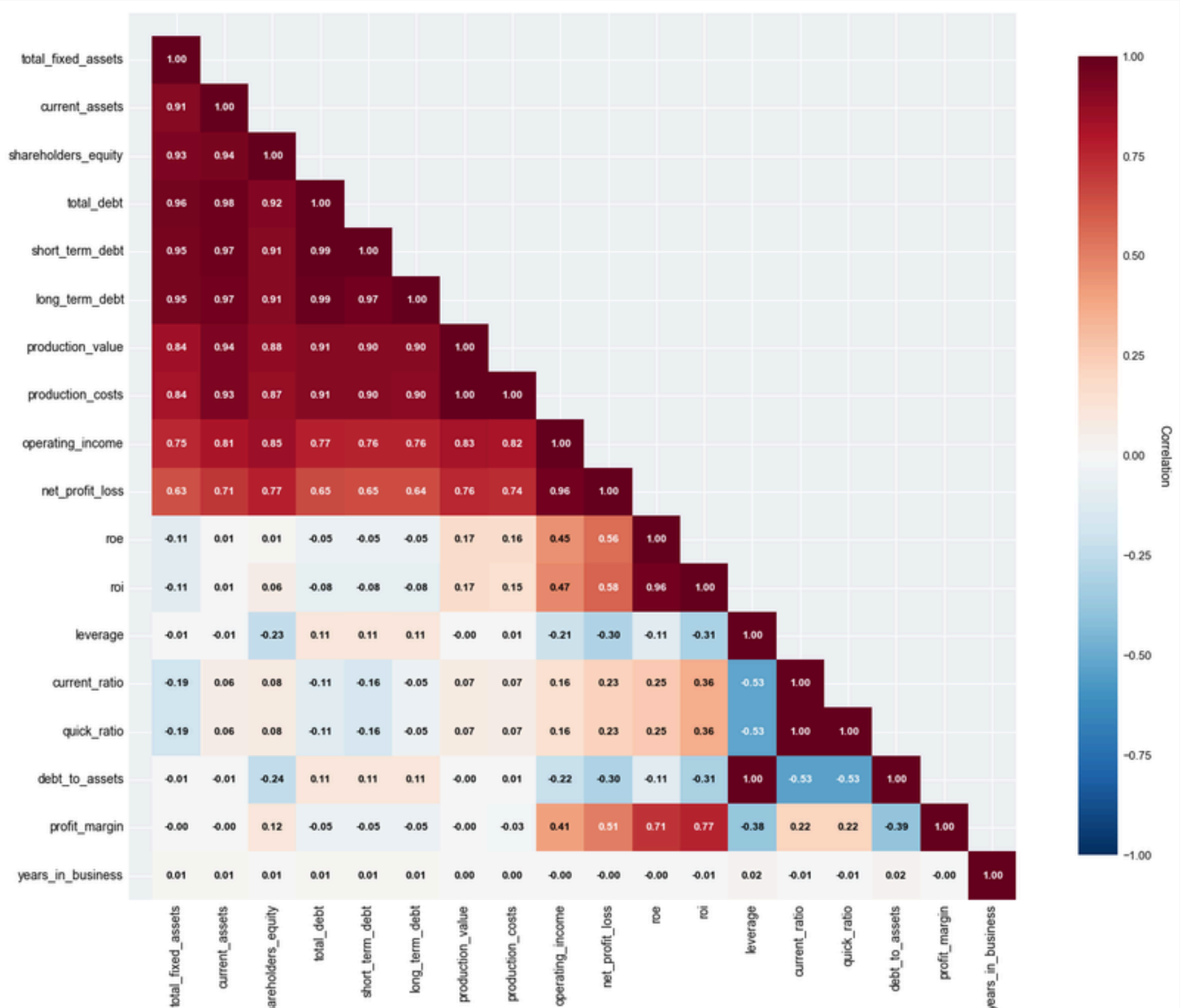


These are **not ordinary missing values**: they encode structural financial distress and should be preserved as a **predictive signal**

MISSING VALUES: TWO MECHANISMS, TWO RESPONSES



FEATURE CORRELATION MATRIX



Key Findings

- Size variables & 3 ratio pairs redundant → **dropped**
- ROE, ROI, profit margin independent → **retained as core features**
- Eliminates mathematical identities (DTA/Leverage) to prevent coefficient inflation, ensuring a parsimonious, audit-ready model

DISTRIBUTION ALIGNMENT & FEATURE STABILITY

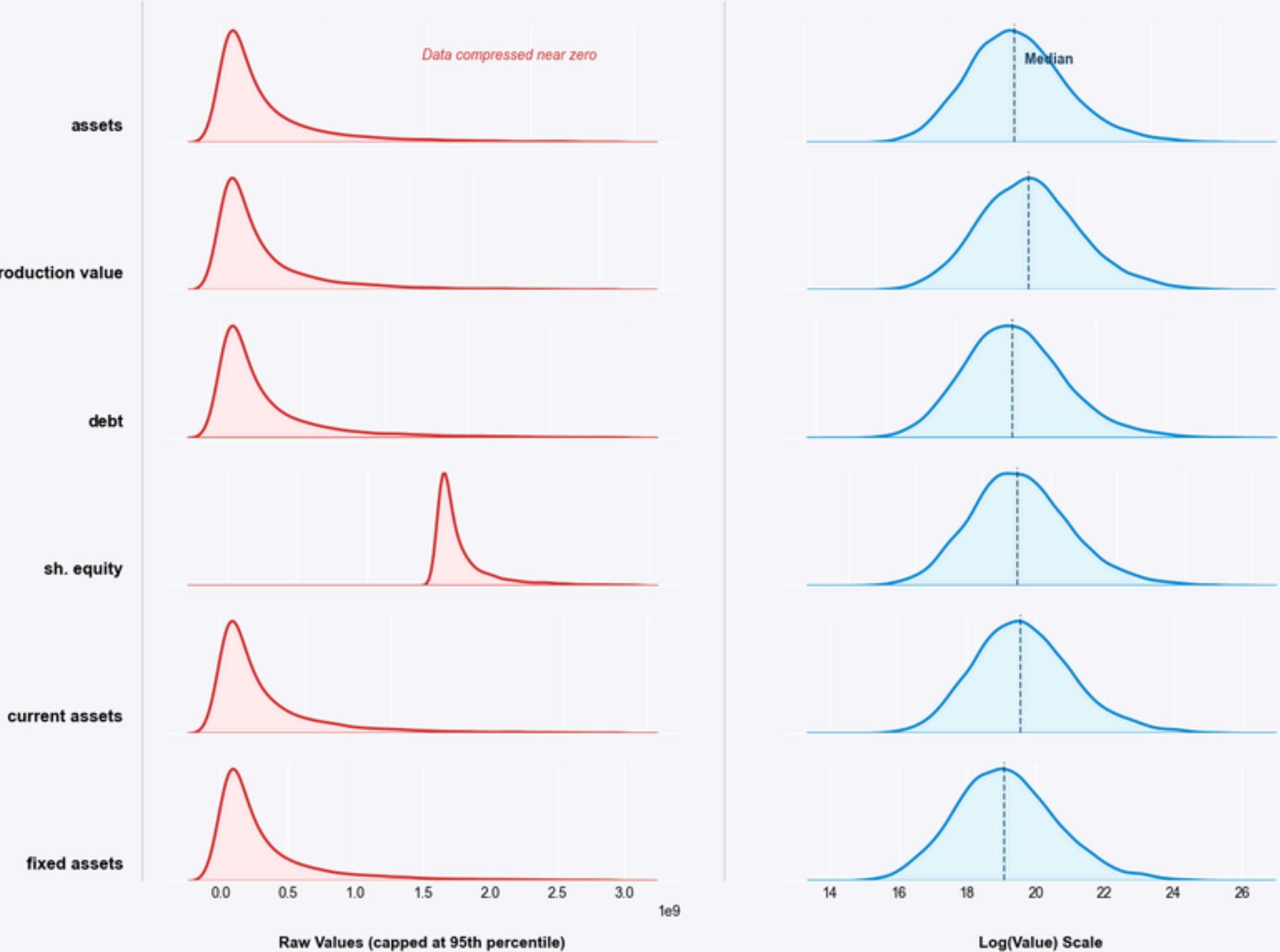
Balance Sheet Variables as Proxies for Company Size

Raw Distribution (Extreme Right Skew)

Data compressed near zero

Log-Transformed (Clear Normal Distribution)

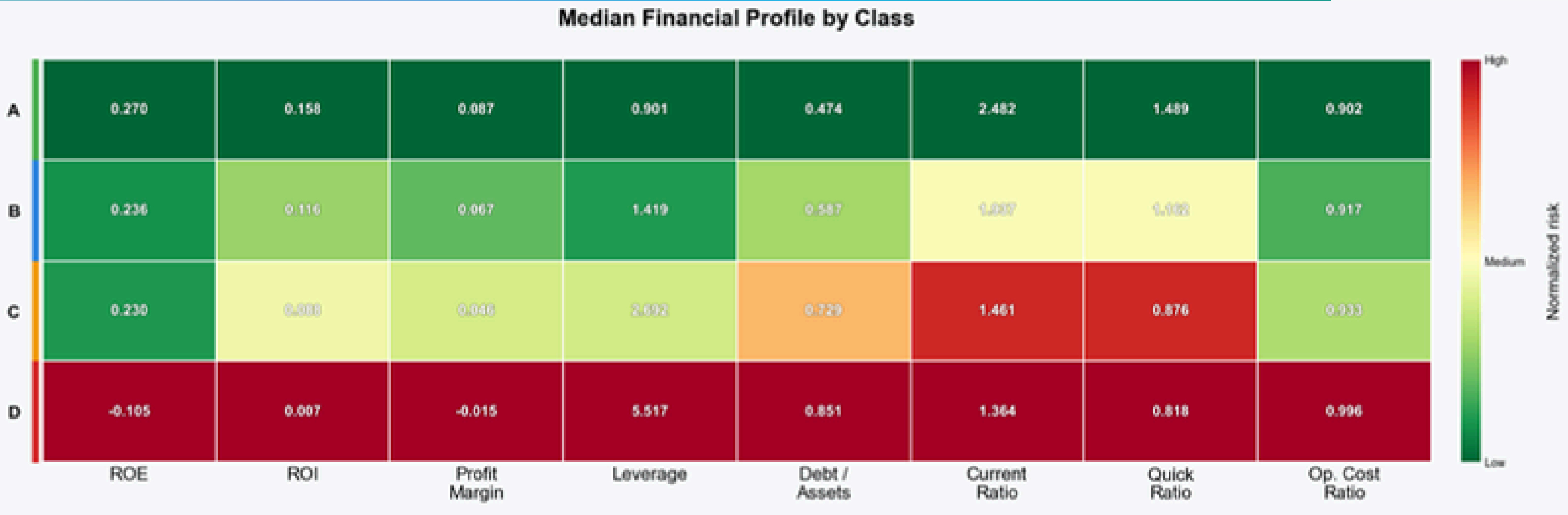
Median



Key Findings

- Raw monetary variables are dropped; only a single log-size proxy is kept

MEDIAN FINANCIAL PROFILE BY HEALTH CLASS



- Ratios worsen consistently **from A → D**, showing a clear **risk gradient**
- Profitability collapses and leverage spike in class D
- Confirms A–D is ordered and ratios matter more than raw size

THANK YOU

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